

Machine Learning Project Proposal: Phase II

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Course: Machine Learning

1. Project Title

CNN-based Classification of Dog Breeds using the Stanford Dogs Dataset.

2. Background and Motivation

Identifying specific dog breeds from images is a classic computer vision challenge. This project aims to automate this process, which has applications in veterinary diagnostics, animal shelter management, and pet-related mobile applications.

3. Problem Statement

The goal is to develop a Convolutional Neural Network (CNN) capable of accurately classifying images into 120 distinct dog breeds despite variations in lighting, pose, and background clutter.

4. Selected Dataset and Dataset Link

Dataset Name: Stanford Dogs Dataset

Link: <http://vision.stanford.edu/aditya86/ImageNetDogs/>

5. Dataset Description

- **Number of images:** 20,580
- **Number of classes:** 120
- **Image type:** RGB
- **Class distribution:** Balanced (approx. 150-200 images per breed)

6. Research Questions

1. How does a custom-built CNN perform compared to a pre-trained model (Transfer Learning) on this specific dataset?
2. What is the impact of image augmentation techniques (rotation, zoom, horizontal flip) on reducing overfitting?

7. Expected Results

I expect the Transfer Learning model (e.g., ResNet50) to achieve an accuracy exceeding 85%, significantly outperforming the baseline custom CNN.

8. Proposed Methodology

Preprocessing involves resizing images to 224x224 pixels and normalizing pixel values. I will use a 70/15/15 split for training, validation, and testing, and apply data augmentation (flips/rotations) to improve generalization.

9. CNN Model Design

Input layer (224x224x3) → 3 blocks of (Conv2D → ReLU → MaxPooling) → Flatten → Dense (128) → Dropout (0.5) → Output (120 nodes/Softmax).

10. Transfer Learning Models

I will use ResNet50, pre-trained on ImageNet. I will freeze the base layers and fine-tune the final classification head for my 120 categories.

11. Evaluation Metrics

Accuracy, Precision, Recall, and F1-Score, along with a Confusion Matrix.

12. Expected Figures and Tables

Training vs. Validation Accuracy and Loss curves, and a confusion matrix to visualize common misclassifications.